

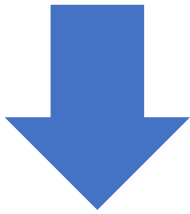


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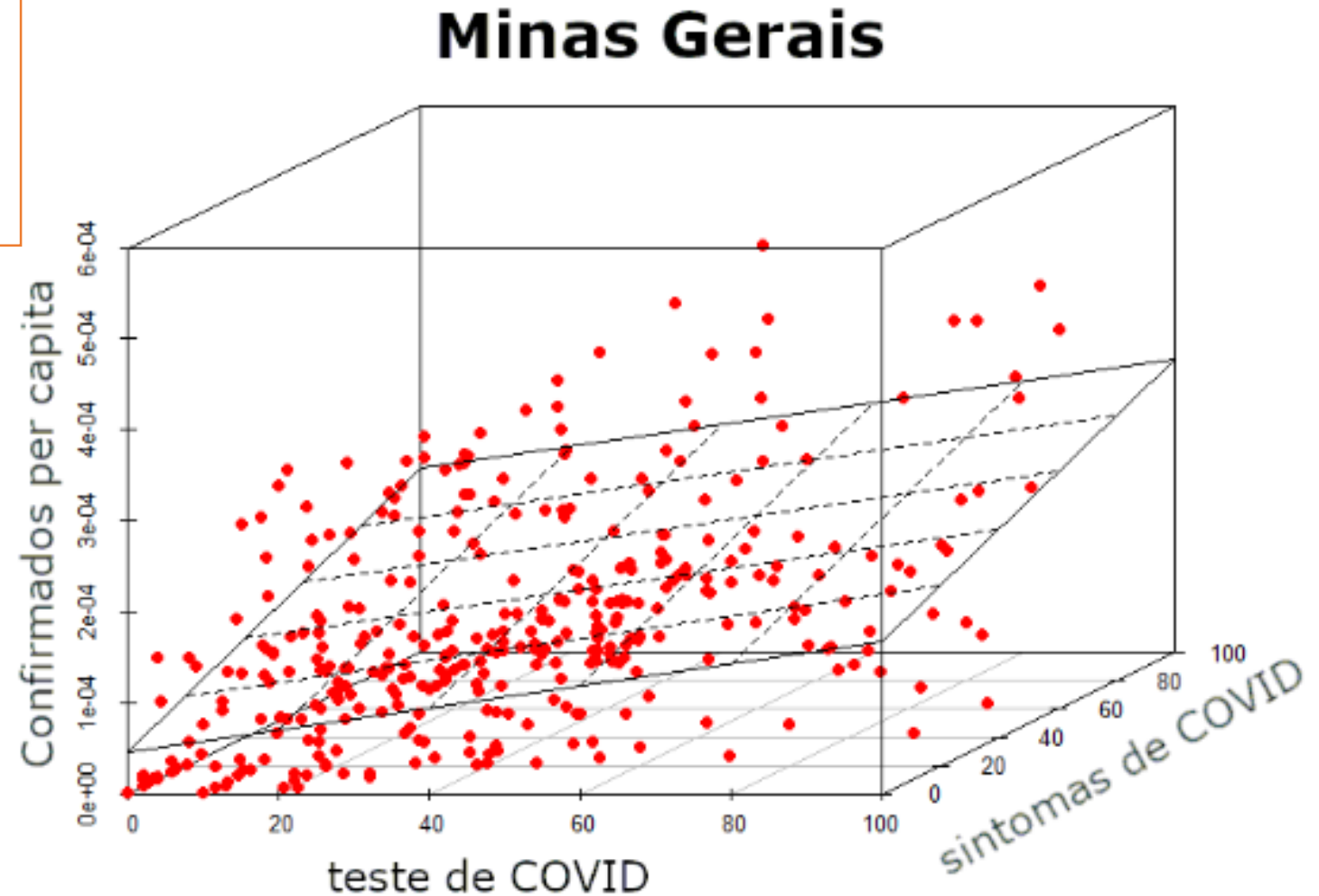
REGRESSÃO LINEAR MÚLTIPLA

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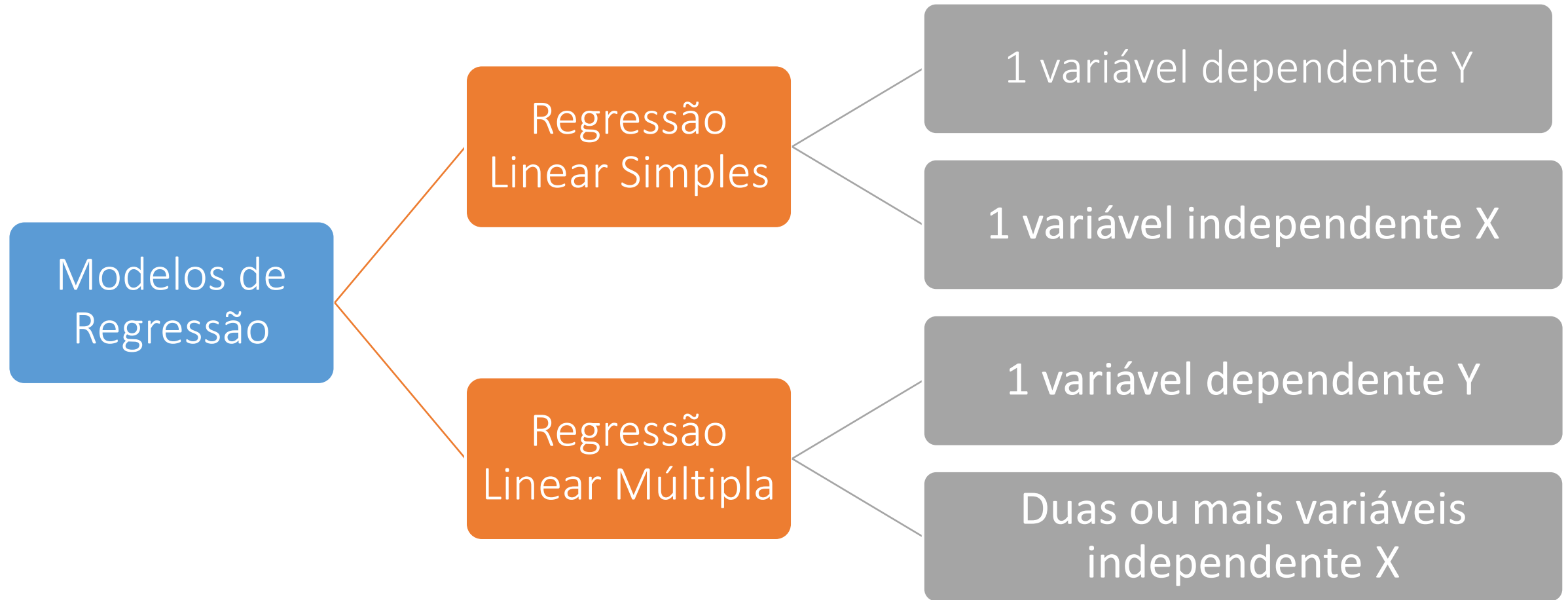
E quando possuímos mais de uma variável independente?



A solução está na :
Regressão Linear Múltipla



REGRESSÃO LINEAR MÚLTIPLA



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The diagram illustrates the components of a multiple linear regression equation. The equation is written as $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i + \hat{\beta}_2 Z_i + \dots + \hat{\beta}_n K_i + e_i$. Each term is color-coded and labeled with arrows:

- $\hat{Y}_i$: Variável Dependente (Dependent Variable)
- $\hat{\beta}_0$: Intercepto Y (Intercept Y)
- $\hat{\beta}_1$: Coeficiente de X (Coefficient of X)
- X_i : Variável independente X (Independent Variable X)
- $\hat{\beta}_2$: Coeficiente de Z (Coefficient of Z)
- Z_i : Variável independente Z (Independent Variable Z)
- $\hat{\beta}_n$: Coeficiente de K (Coefficient of K)
- K_i : Variável independente K (Independent Variable K)
- e_i : Erro Aleatório (Random Error)

Brackets at the bottom group the terms into two main components:

- Componente Linear**: Includes the intercept and all independent variables and their coefficients ($\hat{\beta}_0 + \hat{\beta}_1 X_i + \hat{\beta}_2 Z_i + \dots + \hat{\beta}_n K_i$).
- Componente do Erro Aleatório**: Includes the random error term (e_i).

ESTIMATIVA DOS COEFICIENTES: FORMA MATRICIAL

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i + \hat{\beta}_2 Z_i + \dots + \hat{\beta}_n K_i + e_i$$

$$Y = X B + e$$
$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \\ \vdots \\ \vdots \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_1 & & k_1 \\ 1 & x_2 & & k_2 \\ 1 & x_3 & & k_3 \\ 1 & x_4 & & k_4 \\ 1 & x_5 & \dots & \cdot \\ \vdots & \cdot & & \cdot \\ \vdots & \cdot & & \cdot \\ \vdots & \cdot & & \cdot \\ n & x_n & \dots & k_n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \cdot \\ \cdot \\ \beta_n \end{bmatrix} + \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \\ \cdot \\ \cdot \\ \cdot \\ e_n \end{bmatrix}$$

ESTIMATIVA DOS COEFICIENTES: FORMA MATRICIAL

$$Y = XB + e$$

$$e = Y - XB$$

$$SQR = e'e = (Y - XB)'(Y - XB)$$

$$e'e = \begin{bmatrix} e_1 & e_2 & e_3 & e_4 & e_5 & \cdot & \cdot & \cdot & e_n \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \\ \cdot \\ \cdot \\ \cdot \\ e_n \end{bmatrix} = e^2_1 + e^2_2 + e^2_3 + e^2_4 + e^2_5 + \dots + e^2_n$$

ESTIMATIVA DOS COEFICIENTES: FORMA MATRICIAL

$$\text{SQR} = e'e = (Y - XB)'(Y - XB)$$


$$= Y'Y - 2BX'Y + B'X'XB$$


$$\frac{\partial \text{SQR}}{\partial B} = -2X'Y + 2B'X'X \equiv 0$$

ESTIMATIVA DOS COEFICIENTES: FORMA MATRICIAL

$$-2X'Y + 2B'X'X = 0$$



$$2B'X'X = 2X'Y$$



$$B'X'X = X'Y$$



$$\hat{B} = (X'X)^{-1}X'Y$$



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